

Homework 02 Data Handling, Graphics, More R

Due by 11:59pm, Tuesday 2.3.26

S&DS 2300/5300/ENV 757

Submission Details

This homework assignment contains 1 long problem. Edit this `.Rmd` file and insert your responses under the appropriate problems. Submit the knitted output of this file either as a `.pdf` file.

Your document should look nicely formatted, with R code, plots, and text descriptions nicely integrated.

(1) OK Cupid Dataset This is a dataset first discussed by Kirkegaard and Bjerrekaer. We'll use this dataset several times this semester.

The data itself is stored as a parquet. This is a format where each column is stored separately (rather than as a giant array or matrix). It is more efficient to quickly access and search data. We will create a the usual dataframe.

Run the code below to get the dataset itself and then to get a dataset that is the list of variables in the dataset.

```
#Run the line below one time only
install.packages("arrow")
install.packages("tzdb")
library(arrow)
```

```
##
## Attaching package: 'arrow'

## The following object is masked from 'package:utils':
##
##      timestamp
```

```
#get the data - this takes a moment to load
okcupid <- data.frame(read_parquet("https://github.com/jreuning/sds230_data/raw/refs/heads/main/OKCupid/"))

#get the dataset of questions
okcupidQ <- read_csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/OKCupid/questions.csv")
```

1.1) Write a few lines of code to get a sense of both the dataset and the variables it contains. Include whatever code you use below. Be sure to calculate the size of the dataset. For all code you write, always include a comment line on what you are doing.

As a part of this, you may want to open `okcupidQ` in the Environment (upper right quadrant of RStudio, click on Environment, double click on the dataset.) and browse through the questions. DO NOT INCLUDE OUTPUT AS A RESULT OF USING `head()` or `tail()`.

Write code to get the unique values of `Type` in `okcupidQ`. What do you think these values mean?

Next, write not more than three sentences about what you observe about the dataset. Comment on missing values, the types of questions being asked, and whatever else you might notice.

```
# display head of both datasets in addition to opening the data in environment
# head(okcupid)
# head(okcupidQ)

# understand structure of both
str(okcupid)
```

```
## 'data.frame': 68371 obs. of 2626 variables:
## $ Unnamed..0 : int 1 2 3 4 5 6 7 8 9 10 ...
## $ q2 : chr NA NA NA NA ...
## $ q11 : chr "Horrificed" NA NA NA ...
## $ q12 : chr NA NA NA NA ...
## $ q13 : chr NA NA "No" NA ...
## $ q14 : chr NA NA "No" NA ...
## $ q16 : chr NA NA NA NA ...
## $ q17 : chr "No" NA "No" NA ...
## $ q18 : chr NA NA NA NA ...
## $ q20 : chr NA NA NA NA ...
## $ q22 : chr NA NA NA NA ...
## $ q24 : chr NA NA NA NA ...
## $ q25 : chr NA NA NA NA ...
## $ q26 : chr NA NA NA NA ...
## $ q28 : chr "No" NA "No" NA ...
## $ q29 : chr NA NA NA NA ...
## $ q30 : chr NA NA NA NA ...
## $ q32 : chr "I have little or no interest." NA "I have little or no interest." NA
## $ q33 : chr "Yes" NA NA NA ...
## $ q35 : chr "Love" NA "Love" NA ...
## $ q36 : chr NA NA NA NA ...
## $ q37 : chr NA NA NA NA ...
## $ q38 : chr NA NA NA NA ...
## $ q39 : chr NA NA "No" NA ...
## $ q41 : chr NA NA "Extremely important" NA ...
## $ q42 : chr NA NA NA NA ...
## $ q43 : chr NA NA NA NA ...
## $ q45 : chr NA NA NA NA ...
## $ q46 : chr NA NA "Good" NA ...
## $ q48 : chr NA NA NA NA ...
## $ q49 : chr "Carefree" NA "Carefree" NA ...
## $ q50 : chr NA NA NA NA ...
## $ q55 : chr "No" NA "No" NA ...
## $ q56 : chr "No" NA NA NA ...
## $ q57 : chr NA NA NA NA ...
## $ q60 : chr NA NA "Warm-hearted" NA ...
## $ q61 : chr NA NA NA NA ...
## $ q63 : chr NA NA NA NA ...
## $ q65 : chr NA NA NA NA ...
## $ q67 : chr NA NA NA NA ...
## $ q68 : chr NA NA NA NA ...
## $ q69 : chr NA NA NA NA ...
## $ q70 : chr "No" NA "No" NA ...
```

## \$ q71	: chr "No" NA "No" NA ...
## \$ q73	: chr NA NA "Big city" NA ...
## \$ q74	: chr NA NA "Clothes" NA ...
## \$ q76	: chr NA NA NA NA ...
## \$ q77	: chr "Sometimes" NA "Rarely" "Sometimes" ...
## \$ q78	: chr NA NA NA NA ...
## \$ q79	: chr "Never." NA "I smoked in the past, but no longer." NA ...
## \$ q80	: chr "I never do drugs." NA NA NA ...
## \$ q81	: chr "Rarely" NA "Rarely" NA ...
## \$ q82	: chr NA NA NA NA ...
## \$ q84	: chr NA NA "No" NA ...
## \$ q86	: chr NA NA "No" NA ...
## \$ q87	: chr "No" NA "No" NA ...
## \$ q90	: chr NA NA NA NA ...
## \$ q91	: chr NA NA "Read" NA ...
## \$ q92	: chr NA NA NA NA ...
## \$ q95	: chr NA NA NA NA ...
## \$ q96	: chr NA NA NA NA ...
## \$ q97	: chr NA NA NA NA ...
## \$ q105	: chr "No" NA "No" NA ...
## \$ q106	: chr "No" NA "No" NA ...
## \$ q109	: chr NA NA NA NA ...
## \$ q110	: chr NA NA NA NA ...
## \$ q111	: chr NA NA NA NA ...
## \$ q112	: chr NA NA NA NA ...
## \$ q113	: chr NA NA NA NA ...
## \$ q114	: chr NA NA NA NA ...
## \$ q122	: chr "A lot" NA "A lot" NA ...
## \$ q123	: chr "No" NA "No" NA ...
## \$ q124	: chr NA NA NA NA ...
## \$ q125	: chr NA NA NA NA ...
## \$ q126	: chr NA NA NA NA ...
## \$ q127	: chr NA NA NA NA ...
## \$ q128	: chr "I have no tattoos" NA "I have no tattoos" NA ...
## \$ q129	: chr NA NA "No" NA ...
## \$ q130	: chr NA NA NA NA ...
## \$ q132	: chr NA NA NA NA ...
## \$ q133	: chr NA NA NA NA ...
## \$ q134	: chr NA NA NA NA ...
## \$ q135	: chr NA NA NA NA ...
## \$ q136	: chr "Yes" NA "Yes" NA ...
## \$ q137	: chr "No" NA NA NA ...
## \$ q140	: chr NA NA NA NA ...
## \$ q141	: chr NA NA "Yes" NA ...
## \$ q144	: chr NA NA NA NA ...
## \$ q146	: chr "Kissing in a tent, in the woods" NA "Kissing in Paris" NA ...
## \$ q152	: chr NA NA NA NA ...
## \$ q154	: chr NA NA NA NA ...
## \$ q157	: chr NA NA NA NA ...
## \$ q159	: chr NA NA NA NA ...
## \$ q160	: chr NA NA NA NA ...
## \$ q161	: chr NA NA "More freedom" NA ...
## \$ q163	: chr NA NA NA NA ...
## \$ q165	: chr NA NA NA NA ...

```
## $ q166 : chr NA NA NA NA ...
## $ q170 : chr NA NA NA NA ...
## [list output truncated]
```

```
str(okcupidQ)
```

```
## 'data.frame': 2620 obs. of 10 variables:
## $ X : chr "q2" "q11" "q12" "q13" ...
## $ text : chr "Breast implants?" "How does the idea of being slapped hard in the face during sex
## $ option_1: chr "more cool than pathetic" "Horrificed" "Yes" "Yes" ...
## $ option_2: chr "more pathetic than cool" "Aroused" "No" "No" ...
## $ option_3: chr "" "Nostalgic" "" "" ...
## $ option_4: chr "" "Indifferent" "" "" ...
## $ N : int 24839 28860 22496 32581 31127 8341 38180 21686 16643 31024 ...
## $ Type : chr "N" "N" "O" "O" ...
## $ Order : chr "" "" "" "" ...
## $ Keywords: chr "sex/intimacy; preference; opinion" "sex/intimacy" "sex/intimacy" "sex/intimacy" .
```

```
# get the dimensions of the original dataset - there are 68371 rows and 2626
# columns meaning 68371 responses on potentially 2626 questions
dim(okcupid)
```

```
## [1] 68371 2626
```

```
# dimensions of other dataset - 2620 rows (aka questions) and 10 columns other options or keywords
dim(okcupidQ)
```

```
## [1] 2620 10
```

```
# count n/as
missing_values <- colSums(is.na(okcupid)) # counts NA in each column
sorted_missing_values <- sort(missing_values, decreasing = T) # sorts missing value so most missing is
head(sorted_missing_values, 10) # these are the top 10 columns with missing values
```

```
## q92899 q72000 q155299 q125283 q145537 q116932 q128112 q91348 q118236 q91207
## 68270 68252 68251 68245 68223 68218 68213 68209 68206 68197
```

```
# unique values of Type - N, M, O, blank
unique(okcupidQ$Type)
```

```
## [1] "N" "O" "M" ""
```

```
# return a row of each value to investigate further
blank_type <- okcupidQ[okcupidQ$Type == "", ] # ""
N_type <- okcupidQ[okcupidQ$Type == "N", ] # "N"
M_type <- okcupidQ[okcupidQ$Type == "M", ] # "M"
O_type <- okcupidQ[okcupidQ$Type == "O", ] # "O"

okcupidQ[okcupidQ$X == "p_drug", ] # ""
```

```
##          X          text option_1 option_2 option_3 option_4      N Type Order
## 2544 p_drug Drug-friendly                                     4481
##      Keywords
## 2544
```

```
okcupidQ[okcupidQ$X == "q2", ] # N
```

```
##      X          text          option_1          option_2 option_3
## 1 q2 Breast implants? more cool than pathetic more pathetic than cool
##      option_4      N Type Order          Keywords
## 1          24839      N          sex/intimacy; preference; opinion
```

```
okcupidQ[okcupidQ$X == "q78", ] # M
```

```
##      X          text option_1 option_2
## 56 q78 Generally, do you enjoy being drunk?      Yes      No
##          option_3 option_4      N Type Order
## 56 I don't know, because I've never been drunk. 30487      M      [3]
##      Keywords
## 56 preference
```

```
okcupidQ[okcupidQ$X == "q12", ] # O
```

```
##      X          text
## 3 q12 Divide your age by 2. Have you had sex with at least that many people?
##      option_1 option_2 option_3 option_4      N Type Order      Keywords
## 3      Yes      No          22496      0          sex/intimacy
```

Type is classified as either N, O, M or blank. I think the O is when an answer can be yes or no. N is when there are differing answers a respondent can choose from. M is when there are different answers they can choose from and they are more of a scalar nature. And blank is not for questions but rather for profile/classification variables collected from the user. The dataset is very long and very interesting. It is dating-natured questions asked to respondents to build their profile that was scraped and parsed from OkCupid. Lots of people did not answer many of the questions so there are lots of missing values. It is a very long dataset with a lot of questions that were presented to respondents that can be analyzed,

1.2) Write exactly three questions that you are curious to answer using this dataset.

What are most answered questions. What is the least answered question. How many people, if any, completed the entire survey.

1.3) We are going to calculate a numeric value of income. I did this in the code below.

```
grep("income", names(okcupid)) # this looks for the word income in the column names of okcupid; this al
```

```
## [1] 1469
```

```
okcupid[, 1469][1:100] # this returns the first 100 items of column 1469; this gives an idea of the da
```

```
## [1] NA      NA      NA
## [4] NA      NA      NA
```

```
## [7] NA NA NA
## [10] NA NA NA
## [13] NA NA NA
## [16] "$20,000-$30,000" NA NA
## [19] NA NA NA
## [22] NA "$50,000-$60,000" NA
## [25] NA NA NA
## [28] NA NA NA
## [31] "$30,000-$40,000" "$40,000-$50,000" NA
## [34] NA NA NA
## [37] NA NA NA
## [40] NA NA NA
## [43] NA "$20,000-$30,000" "$80,000-$100,000"
## [46] NA "Rather not say" NA
## [49] NA NA NA
## [52] NA NA NA
## [55] NA NA NA
## [58] NA NA "Less than $20,000"
## [61] NA NA NA
## [64] NA NA NA
## [67] NA NA NA
## [70] NA NA NA
## [73] NA NA NA
## [76] NA NA NA
## [79] NA NA NA
## [82] NA NA NA
## [85] NA "Rather not say" NA
## [88] NA NA "Rather not say"
## [91] NA "$30,000-$40,000" NA
## [94] "Rather not say" NA NA
## [97] NA NA NA
## [100] NA
```

```
names(okcupid)[1469] # this returns the name of the 1469th column of okcupid; it is d_income, so you can
```

```
## [1] "d_income"
```

```
income <- okcupid[, 1469] # this creates a variable called income that stores the data from all rows of
```

```
unique(income) # this returns the unique values; allowing you to see what the range and answer options
```

```
## [1] NA "$20,000-$30,000" "$50,000-$60,000"
## [4] "$30,000-$40,000" "$40,000-$50,000" "$80,000-$100,000"
## [7] "Rather not say" "Less than $20,000" "$60,000-$70,000"
## [10] "More than $1,000,000" "$100,000-$150,000" "$70,000-$80,000"
## [13] "$250,000-$500,000" "$150,000-$250,000" "$500,000-$1,000,000"
```

```
parse_income <- function(x) { # this is creating a new function
  if (is.na(x) || x == "Rather not say") { # if it has an na or says rather not say,
    return(NA_real_) # returns NA
  } else if (x == "More than $1,000,000") { # if the value is "more than 1,000,000
    return(1000000) # return the number 1,000,000
  }
}
```

```

} else if (x == "Less than $20,000") { # if the value is less than $20,000
  return(20000) # return the number 20,000
} else if (grepl("-", x)) { # otherwise, see if the elements have -
  nums <- gsub("[\\$,]", "", x) # if it has a $ or ,, it replaces that character with a space and stores it in nums
  parts <- strsplit(nums, "-")[1] # this splits anything in nums at the - and stores the entire vector in parts
  return(mean(as.numeric(parts))) # this returns the mean of the parts as a numeric vector
}
return(NA_real_) # if nothing matches the criteria above, return NA_real_
}

# this is looking through the income variable and parsing it so it is easier to use and understand. if

okcupid$income_numeric <- as.numeric(sapply(income, parse_income))# a column is added to okcupid that is income_numeric
# we can then see in the income_numeric column of okcupid there is only 1 value in each row: NA, 1,000,000

```

Admittedly, much of the code above is things we have not yet discussed.

Run the code to see what it does. You may want to run individual lines or subsections. Feel free to ask AI specific questions about functions.

Your main job is to add comments to the code to describe what each part does. Furthermore, in your comments indicate the thought process behind the code (i.e. what question prompted the code?).

Putting the code into AI will do this for you - that said, you'll ultimately want to make sure that if I or others gave you the code without comments that you could read the code for yourself.

Finally - make any plot you like to examine the distribution of income and discuss what you observe.

```

# look at how many unique numbers there are in income to determine range of graph
unique(okcupid$income_numeric)

```

```

## [1] NA 25000 55000 35000 45000 90000 20000 65000 1000000
## [10] 125000 75000 375000 200000 750000

```

```

sort(unique(okcupid$income_numeric))

```

```

## [1] 20000 25000 35000 45000 55000 65000 75000 90000 125000
## [10] 200000 375000 750000 1000000

```

```

# there are 14 unique values, including NA, spanning 20,000 - 1,000,000
# there is basically a $10,000 increase with each but it's not steady every increase
# make a table to see how many frequencies are in each answer
# this will help me check my graph is showing what is actually in the variable
table_income_numeric <- (table(okcupid$income_numeric))
table_income_numeric

```

```

##
## 20000 25000 35000 45000 55000 65000 75000 90000 125000 2e+05 375000
## 2819 2480 2241 1805 1359 832 671 891 920 340 121
## 750000 1e+06
## 36 422

```

```
# see how many are na
sum_na_income_numeric <- sum(is.na(okcupid$income_numeric))
sum_na_income_numeric
```

```
## [1] 53434
```

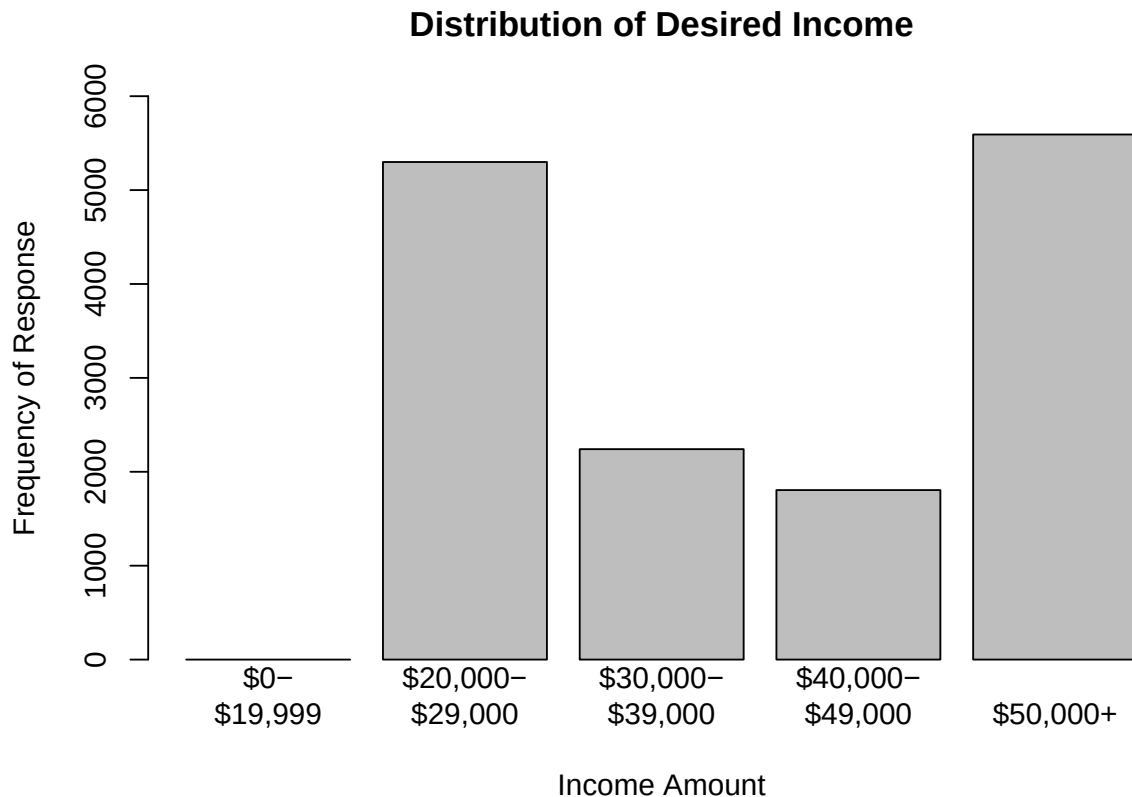
```
# I want to keep the N/As to show how many people did not answer or preferred not to say but 53,000 is
# this will skew th appearance of the rest of the values so I am going to drop them
# If I make a barplot I need to make specific bins rather than what is the default
```

```
# Create a new variable of the bins
income_bins <- cut(
  okcupid$income_numeric,
  breaks = c(0, 20000, 30000, 40000, 50000, 1000000), # cuts the data at these ranges making 5 bins
  right = FALSE, # left end included, right end excluded ex: 0-19,999
  include.lowest = TRUE, # very lowest value is included
  labels = c("$0-\n$19,999", "$20,000-\n$29,000", "$30,000-\n$39,000", "$40,000-\n$49,000", "$50,000+")
)
```

```
# see how many responses are in each level with new binning
table(income_bins)
```

```
## income_bins
##      $0-\n$19,999 $20,000-\n$29,000 $30,000-\n$39,000 $40,000-\n$49,000
##                0                5299                2241                1805
##      $50,000+
##      5592
```

```
# plot the barplot
barplot(table(income_bins),
  main = "Distribution of Desired Income",
  ylab = "Frequency of Response",
  xlab = "Income Amount",
  ylim = c(0,6000))
```

The majority of respondents (53,434) either did not answer or chose ‘prefer not to say.’ Of respondents who gave an answer, the majority of respondents answered over \$50,000 income. The next most frequent group was \$20,000-29,000. Both \$30,000-39,000 and \$40,000-49,000 had less than 3000 respondents. Very, very few respondents said less than \$20,000.

(1.4) Pick any variable EXCEPT q30458 that starts with q that has four possible outcomes (options).

- Create and display a table called `table1` that shows how many of instances there were of each level.
- Write a line that calculates the percentage of values that were from the fourth level of the question, rounded to the nearest whole percent.
- Write a line of code to use `paste()` to make a statement like “x% of respondents responded (response) to question (text of the question pulled from okcupidQ).”

```
# I need to pull this question line from okcupidQ to get the text of the question and responses
okcupidQ[okcupidQ$X == "q54", ]
```

```
##      X                text option_1 option_2 option_3
## 38 q54 Which, on you, is closest to perfection? Your mind Your body Your wallet
##      option_4      N Type Order                Keywords
## 38 Your soul 22704      N      religion/superstition; descriptive
```

```
# the fourth option or level is 'Your soul'
```

```
# create table of q54
(table1 <- table(okcupid$q54))
```

```
##
## Your body Your mind Your soul Your wallet
## 933 10516 10851 404

# create percent that responded to 4th option
percent_4 <- round(table1["Your soul"]/sum(table1) *100)
percent_4
```

```
## Your soul
## 48
```

```
# return text aka question from row
question_q54 <- okcupidQ[okcupidQ$X == "q54", "text"] # returns as character

# return 4th option from row
response_q54 <- okcupidQ[okcupidQ$X == "q54", "option_4"]

paste(percent_4, "% of respondant responded", response_q54, "to question:", question_q54)
```

```
## [1] "48 % of respondant responded Your soul to question: Which, on you, is closest to perfection?"
```

(1.5) Create a barplot using the `barplot()` function for whatever question you chose in part (1.4).

- The labels of the barplot should also contain the whole number percentages for each possible outcome.
- Format your graph, and make sure the bars are horizontal.
- Change the margins using `par(mar = ...)` if your labels are being cut off, and change the character size if necessary.

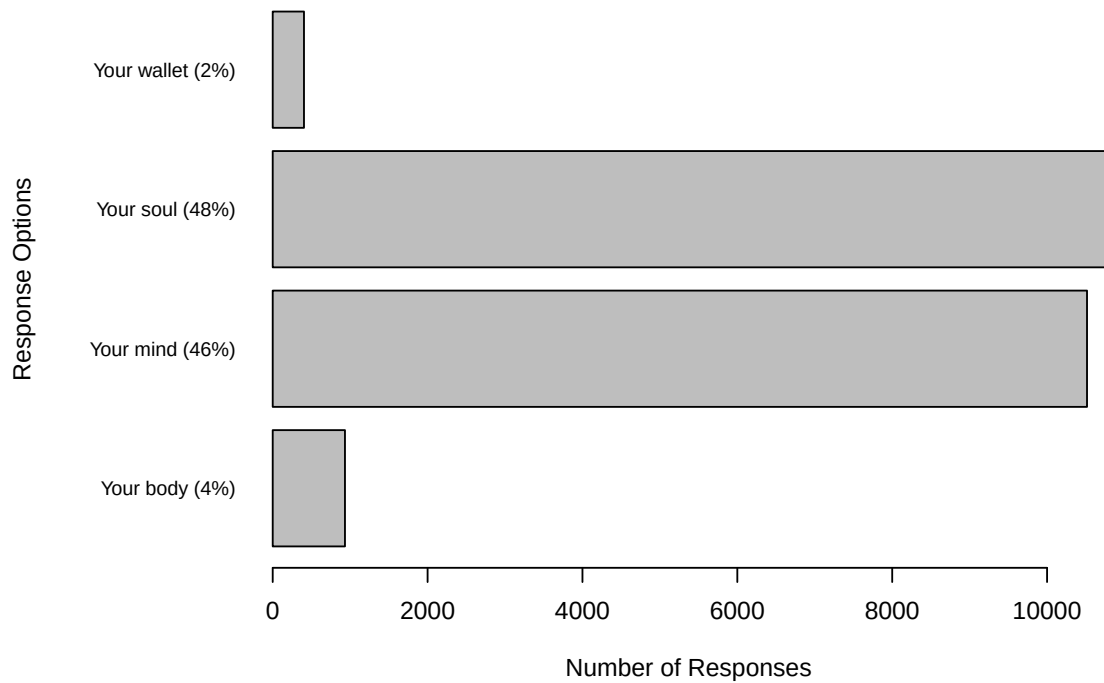
```
#create percentages and add to names
percents_all <- round(table1/sum(table1) *100) # creates percents of all levels
names(table1) <- paste0(names(table1), " (",percents_all,"%)" ) # adds (%) after each name for graphing

# adjust margin to fit labels
par(mar = c(5, 12, 6, 2), cex = 0.8) # bottom, left, top, right; character size

# graph bargraph
barplot((table1),
  horiz = T, # makes bars horizontal
  main = "Response to Question 54 on OkCupid",
  xlab = "Number of Responses",
  las = 1,
  cex.names = 0.8)

mtext("Response Options", side = 2, line = 8, cex = 0.8) # adds the y axis label after so it's not on t
```

Response to Question 54 on OkCupid



(1.6) Choose any variable that has a Yes/No response.

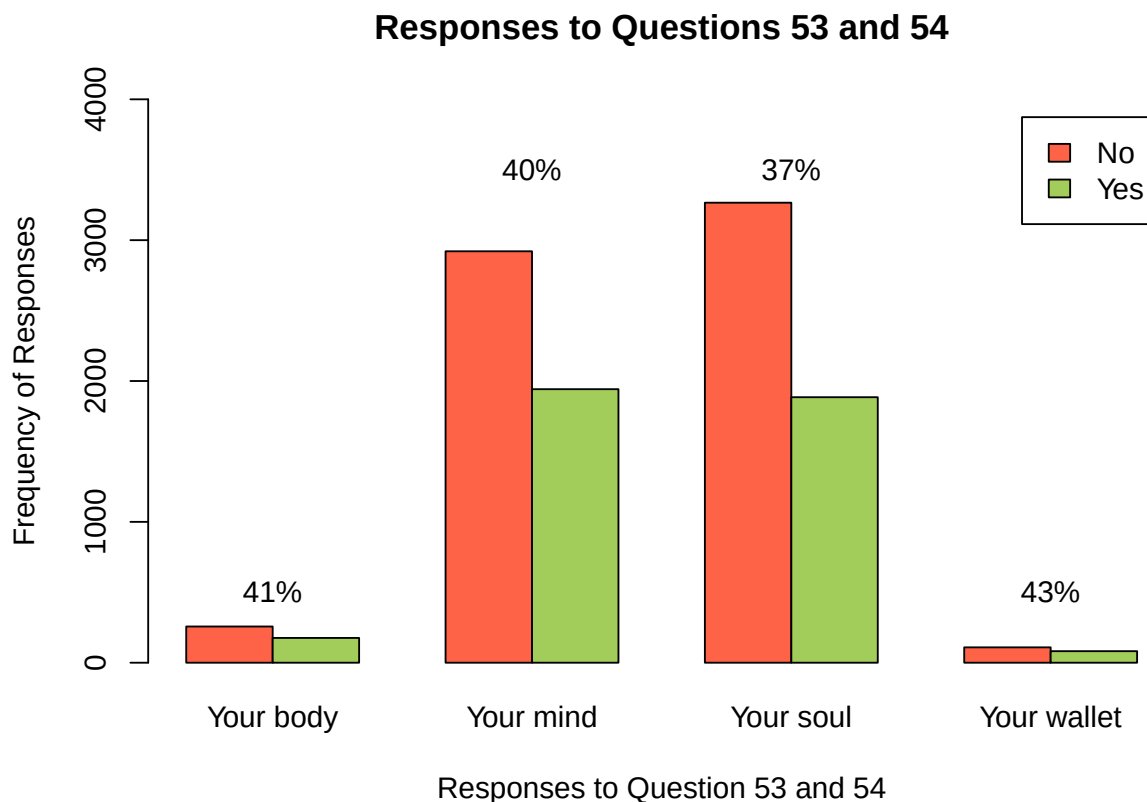
- Make a two-way table between this new question and your question in 1.4.
- Calculate the percent of people for each level of your 1.4 question that answered 'Yes' to your new question.
- Following the example in Class 4, create a barplot akin to that in lines 431-442 with similar percentages displayed. This will take some adjusting. You can make horizontal or vertical. Make sure no labels are cut off.
- Does it seem like these variables might be related?

```
# two way table of q53 and q54
two_questions <- table(okcupid$q53, okcupid$q54)
two_questions
```

```
##
##      Your body Your mind Your soul Your wallet
## No         257    2921    3266         109
## Yes        176    1942    1885          82
```

```
# calculate percent of people for each response that answered yes
# numerator = yes to q53/denominator = everyone who answered that level aka the column total
two_questions_label <- paste(round(two_questions[2,]/apply(two_questions, 2, sum)*100), "%", sep = " ")
```

```
# create barplot
barplot((two_questions),
  beside = T, # makes besides not stacked
  xlab = "Responses to Question 53 and 54",
  ylab = "Frequency of Responses",
  main = "Responses to Questions 53 and 54",
  col = c("tomato", "darkolivegreen3"),
  legend = T, # adds legend
  ylim = c(0, 4000)) # extends y limit to 3500 to see full responses
text(c(2, 5, 8, 11), c(500, 3500, 3500, 500), two_questions_label)
```



I looked at question 53 which asked “when telling stories do you like to exaggerate” (yes/no) and question 54 which asked “Which, on you, is closest to perfection?” (Your body, your mind, your soul, or your wallet). It does not appear these are connected, although it does seem more people do not like to exaggerate when telling stories than people who do as more people answered no in every category. Additionally, more people felt like their mind and soul were closer to perfection than their wallet or body. But there did not seem to be any relatedness between the questions.

(1.7) Choose any two variables that start with the letter p. These are questions that are calculated scales.

- Calculate summary statistics using `summary()` and also the standard deviation for each variable (rounded to 2 decimal places).
- Make histograms for each variable. Format your histogram.

- Describe the shape of each histogram (use words like ‘symmetric’ or ‘skewed’) and discuss what you observe.

```
# summary of each
```

```
summary(okcupid$p_exer)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## -94.00  -30.00   -3.00    1.71   40.00   97.00   49948
```

```
summary(okcupid$p_conf)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##       2.00   15.00   32.00   34.82   50.00   98.00   56268
```

```
# standard deviation of each rounded to 2 decimal places
```

```
round(sd(okcupid$p_exer, na.rm = TRUE), 2) # na.rm=TRUE means NA should be removed
```

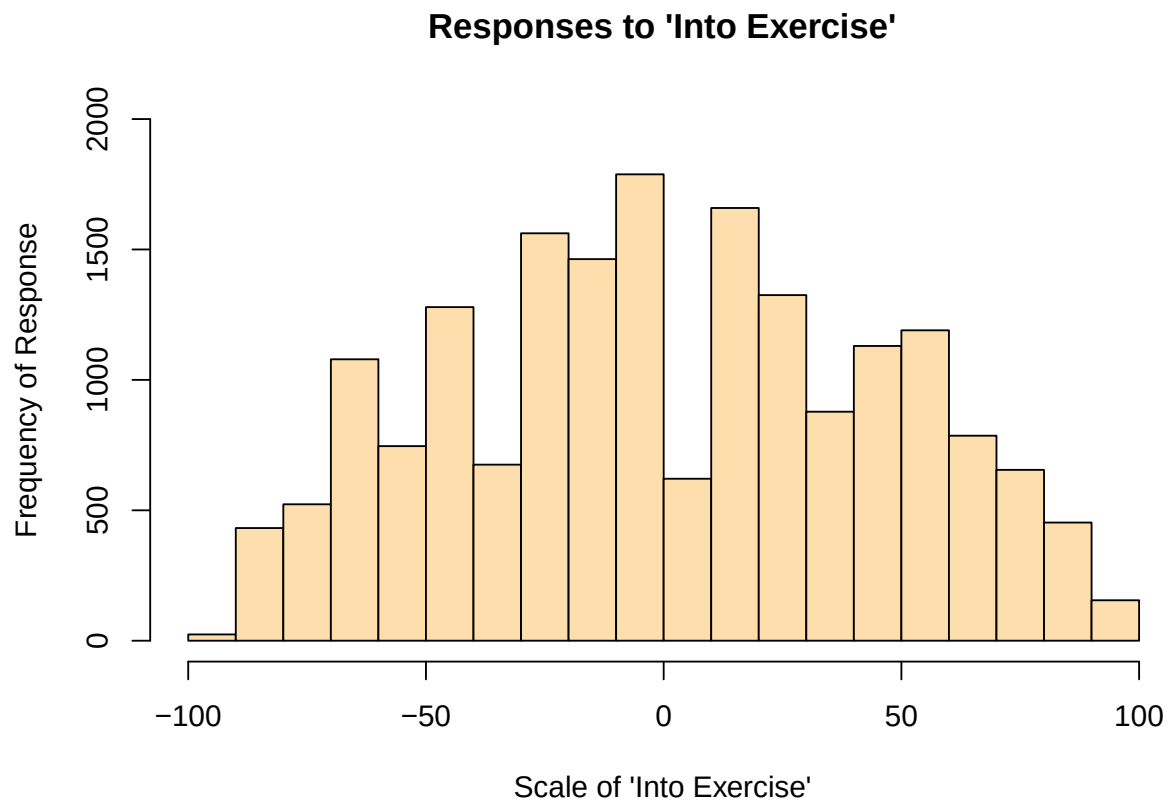
```
## [1] 45.43
```

```
round(sd(okcupid$p_conf, na.rm = TRUE), 2)
```

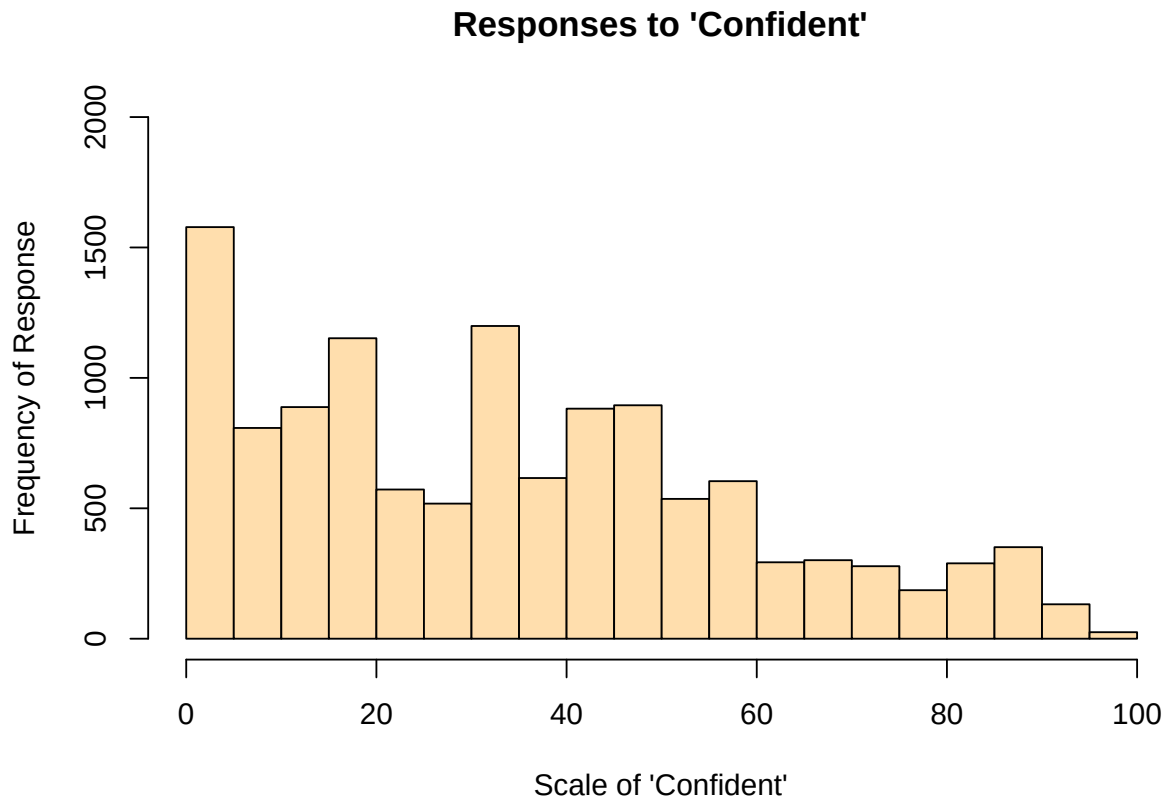
```
## [1] 24.35
```

```
# histogram for Into Exercise
```

```
exercise_histogram <- hist((okcupid$p_exer),
  xlab = "Scale of 'Into Exercise'",
  ylab = "Frequency of Response",
  main = "Responses to 'Into Exercise'",
  ylim = c(0, 2000),
  col = "navajowhite")
```



```
# histogram for Confident
confidence_histogram <- hist((okcupid$p_conf),
  xlab = "Scale of 'Confident'",
  ylab = "Frequency of Response",
  main = "Responses to 'Confident'",
  ylim = c(0, 2000),
  col = "navajowhite")
```



The responses to 'Into Exercise' are much more normal or symmetric in that the majority of the answers are in the middle range, but it has a very wide range with all bins having at least one respondent. The responses to 'Confident' are skewed to the right. More people identify with being low on the scale of confidence.

(1.8) Choose either of the variables you used in 1.7 AND the Yes/No variable you choose in 1.6.

- Get summary statistics for the p_ variable by each level of your Yes/No variable. Compare the medians and means of these two distributions - what do you observe?

```
tapply(okcupid$p_conf, okcupid$q53, summary)
```

```
## $No
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##    2.0   14.0   26.0   29.4   43.0   96.0   5907
##
## $Yes
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##    2.00  15.00  31.00  32.22  48.00  93.00   3616
```

Of people who said no, they do not like to exaggerate when telling stories, they had a lower mean and median score of confidence compared to people who did like to exaggerate when telling stories. Within those who answered no, the difference in between mean and median was 3 and within those who answered yes, the difference between mean and median was 1. Both had a similar spread and quartile breakdown meaning the majority of data lied within similar ranges although the distribution was not similar.

(1.9) Make a scatterplot of your two p_ variables from from 1.7).

- Set a non-default plot character.
- Create the plot using different colors for each level of your binary variable from 1.6.
- Add a legend to your plot
- Format and label your plot
- Discuss any trends you see as well as any differences between groups.

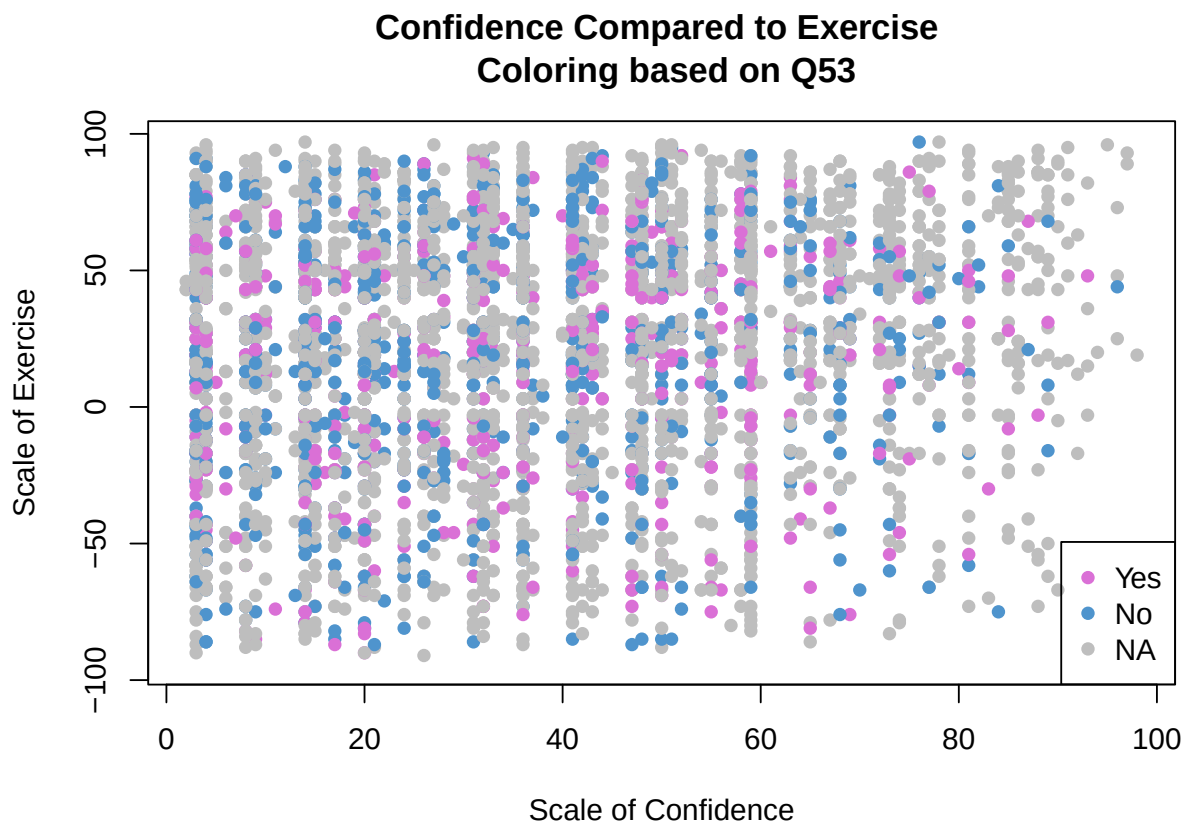
```
# map options of q53 to colors
# find levels: NA, yes, no
unique(okcupid$q53)
```

```
## [1] NA      "No"  "Yes"
```

```
colors_q53 <- ifelse(
  is.na(okcupid$q53), "gray",
  ifelse(okcupid$q53 == "Yes", "orchid", "steelblue3")
)
```

```
# plot for p_conf compared to p_exer and based on q53 (binary)
confidence_plot <- plot(okcupid$p_conf, okcupid$p_exer,
  pch = 16,
  col = colors_q53,
  xlab = "Scale of Confidence",
  ylab = "Scale of Exercise",
  main = "Confidence Compared to Exercise \n Coloring based on Q53")

legend(
  "bottomright",
  legend = c("Yes", "No", "NA"),
  col = c("orchid", "steelblue3", "gray"),
  pch = 16
)
```

I wanted to try one without the NAs to see if there is more of a relationship between the answered one.

```
# drop na
drop_na_q53 <- na.omit(okcupid$q53)
```

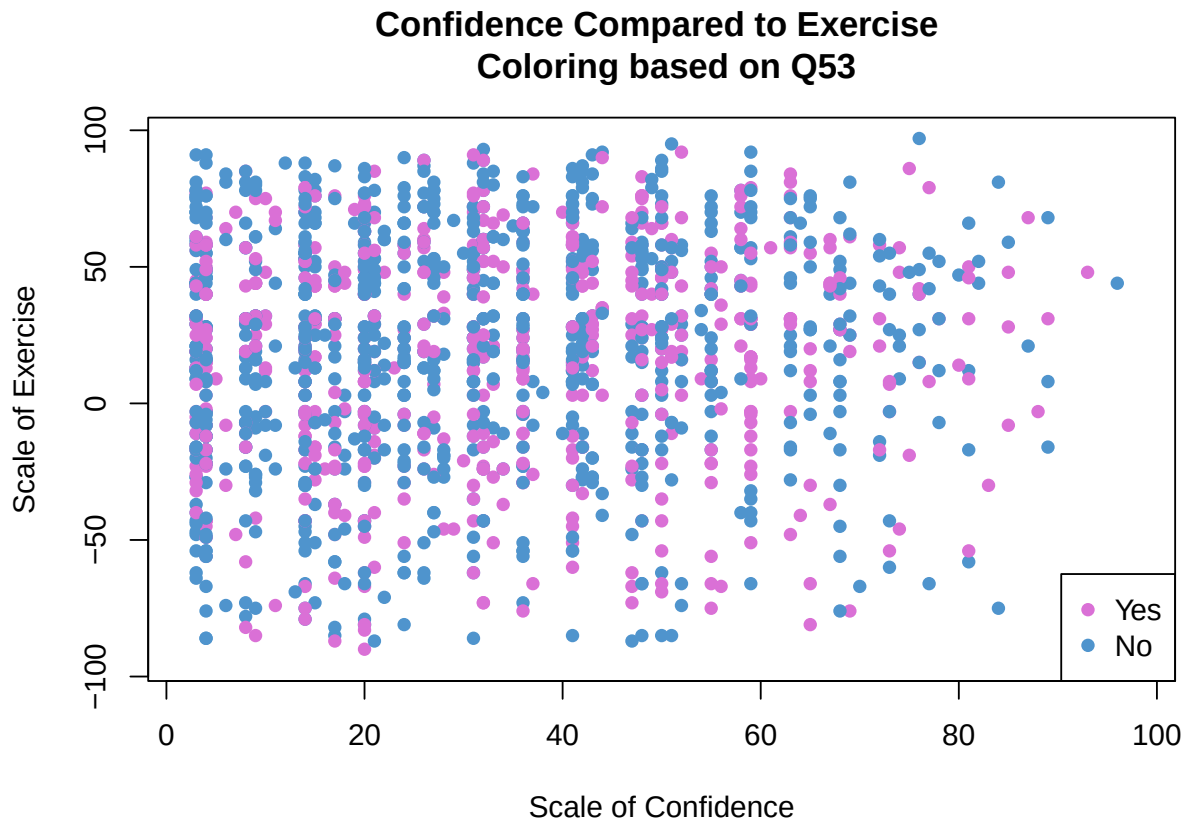
```
# ensure na dropped
unique(drop_na_q53)
```

```
## [1] "No" "Yes"
```

```
# color yes and no onlys
colors_q53_na <- ifelse(okcupid$q53 == "Yes", "orchid", "steelblue3")
```

```
# plot again
confidence_plot_na <- plot(okcupid$p_conf, okcupid$p_exer,
  pch = 16,
  col = colors_q53_na,
  xlab = "Scale of Confidence",
  ylab = "Scale of Exercise",
  main = "Confidence Compared to Exercise \n Coloring based on Q53")
legend(
  "bottomright",
  legend = c("Yes", "No"),
  col = c("orchid", "steelblue3"),
```

```
pch = 16
)
```



I don't see any trends from the graphs. Very similar to what was seen in the bargraph, they don't really have any relationship to one another but we do see points in almost all areas of the graph meaning all answers were represented. The exercise data was more normally spread and represented across the y axis. More of the confidence data is on the lower side of the scale, or the x axis, with very few at the higher end of the x axis (skewed right). Taking out the NAs did not create any noticeable trends either.

(1.10) Create a subset of the original dataset that meets the following conditions:

- It includes ONLY the following variables: all five of the variables you have examined so far (income_num, your four level variable, your two p_ indices, and your binary variable) as well as the variable d_astrology_sign)
- It has complete data for the variables listed above
- You only want people who have Ca or Sa as the first two letters of their astrological sign.

Use both the %in% operator and the substr() function to complete this task. You'll need to look up information about how these work.

Get the dimension of your final dataset. Comment on the reduction in size from the original dataset.

```
# create subset of the 5 variables: q53, q54, p_conf, p_exer, income_numeric)
subset_data <- subset(okcupid, select = c("q53", "q54", "p_conf", "p_exer", "income_numeric", "d_astrol"))

# check dimensions here
dim(subset_data) # 68371 rows
```

```
## [1] 68371      6
```

```
# drop any nas
subset_data <- na.omit(subset_data)

# check dimensions here
dim(subset_data) # 453 rows
```

```
## [1] 453      6
```

```
# only choose astrological signs
subset_data <- subset_data[substr(subset_data$d_astrology_sign, 1, 2) %in% c("Ca", "Sa"), ]

# check dimensions
dim(subset_data) # 105 rows
```

```
## [1] 105      6
```

Isolated to only the 5 wanted variables, there were 68,371 rows/responses. After dropping the NAs to isolate only complete data, there was 453 rows/responses. After choosing only astrological signs that start with Ca or Sa, there were 105 rows/responses. This is a drastic reduction from where we started and highlights how few surveys are truly complete, even when only considering some levels of 5 variables.

(1.11) For any one of the questions you wrote in 1.2), investigate this BRIEFLY with some combination of a plot and basic summary statistics (or a table).

Did anyone complete the full survey.

```
# I need to sum all of the questions aka columns and then sort them and figure out if any are empty.

# drop any rows with na
complete_survey <- na.omit(okcupid)

# get dimensions of all completed survey
dim(complete_survey)
```

```
## [1]      0 2627
```

```
# no one completed the entire survey

# I don't understand how the p variable works so I want to isolate q and see if anyone answered all of

# isolate only the q columns
q_columns <- grep("^q", names(okcupid), value = TRUE) # looks for columns that start with q, returns ma
```

```
# create df of only the q columns
q_df <- okcupid[, q_columns]

# drop NAs
complete_q_df <- na.omit(q_df)

# dimensions of complete q question
dim(complete_q_df)
```

```
## [1] 0 2541
```

```
# 0 people also completed all q questions
```

Because no one answered the complete survey, I didn't think this would make very pretty plots, so I investigated my other questions. *What was the most and least frequently answered question*

```
# sums number of responses per column
answered_counts <- colSums(!is.na(q_df))

# see what was the least frequently answered questions
least_answered_q <- sort(answered_counts)
head(least_answered_q)
```

```
## q92899 q72000 q155299 q125283 q145537 q116932
## 101 119 120 126 148 153
```

```
# go back into cupidQ to see what these columns are
okcupidQ[okcupidQ$X %in% c("q92899", "q72000", "q155299"), ]
```

```
## X
## 2228 q72000
## 2486 q92899
## 2500 q155299
##
## 2228 Which of the following four statements best sums up how you feel about religion / faith when it
## 2486 Would you be inclined to influence a pa
## 2500 Would you be okay with your partner having
## option_1
## 2228 I have a faith, I wish my partner to have one also
## 2486 never,they can't be vanilla enough
## 2500 Yes, they can have as many as they like
## option_2
## 2228 I have a faith, I don't mind if my partner doesn't
## 2486 Not unless s/he was really vanilla
## 2500 Yes, but only if the other computer was a laptop
## option_3
## 2228 I don't have a faith, neither should my partner
## 2486 Probably,unless they were already
## 2500 I don't really mind
## option_4 N Type Order
## 2228 I don't have a faith, I don't mind about my partne 119 0 3, 4, 2, 1
```

```
## 2486          I'd push as far as s/he could go! 101    0
## 2500          No 120    0
##
##                      Keywords
## 2228  religion/superstition; preference
## 2486          sex/intimacy
## 2500  preference; descriptive; technology
```

```
# see what was the most frequently answered questions
most_answered_q <- sort(answered_counts, decreasing = TRUE)
head(most_answered_q)
```

```
## q34113 q64664 q85419 q416235 q20062 q16053
## 68371 68371 68371 68371 68371 58043
```

```
# go back into cupidQ to see what these columns are
okcupidQ[okcupidQ$X %in% c("q34113", "q64664", "q85419"), ]
```

```
##          X
## 1359 q34113
## 2142 q64664
## 2419 q85419
##
## 1359          How do you feel about government-subsidized food programs (free lunch, food stamps
## 2142 Do you think it is okay to open old graves to get more knowledge of ancient cultures and their l
## 2419          Which type of wine would you prefer to drink outside of a meal, such as for l
##
##                      option_1
## 1359          No problem
## 2142          Yes.
## 2419 White (such as Chardonnay, Riesling).
##
##                      option_2                      option_3
## 1359          It's okay, if it is not abused    Okay for short amounts of time
## 2142          No.                                I don't really care.
## 2419 Red (such as Merlot, Cabernet, Shiraz). Rosé (such as White Zinfandel).
##
##                      option_4    N Type    Order    Keywords
## 1359    Never - Get a job 31769    0          politics
## 2142          1148          0 2, 3, 1    opinion
## 2419 I don't drink wine. 18838    N          descriptive
```

```
# table of the question and the question number
question_table <- okcupidQ[okcupidQ$X %in% c("q92899", "q72000", "q155299", "q34113", "q64664", "q85419",
```

```
# add the response numbers as well
question_table$Responses <- answered_counts[question_table$X]
```

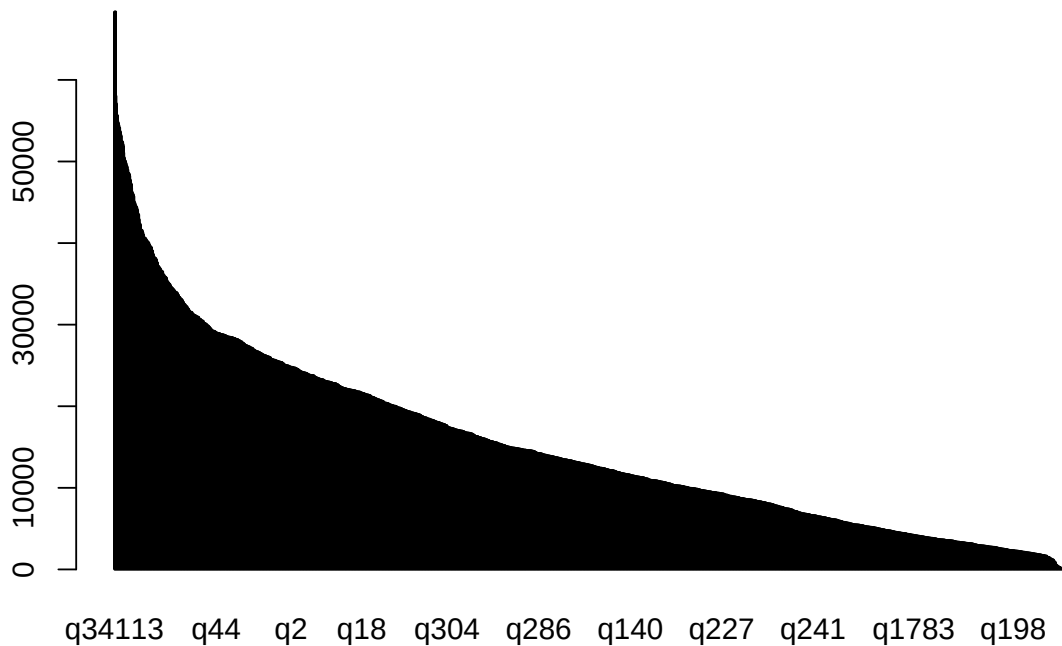
```
# sort so table is decreasing by response number rather than index
question_table <- question_table[order(question_table$Responses, decreasing = TRUE), ]
```

```
# display final table
question_table
```

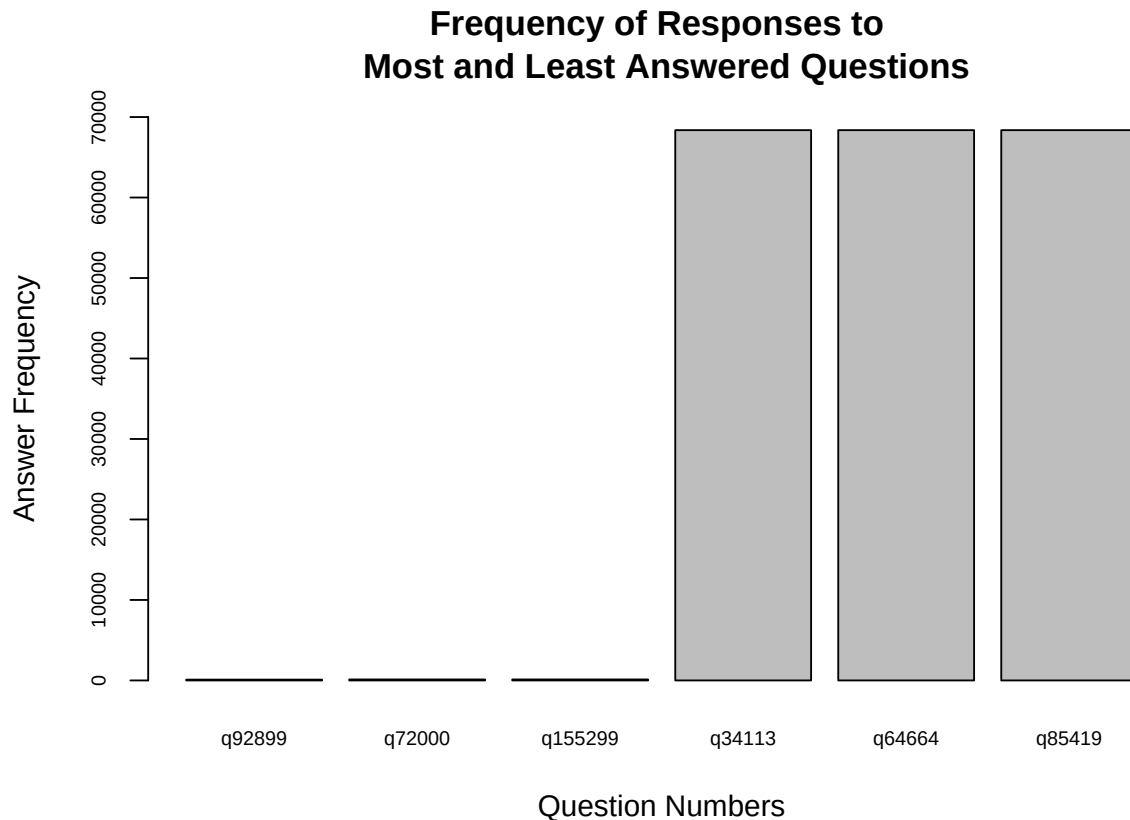
```
##          X
## 1359 q34113
```

```
## 2142 q64664
## 2419 q85419
## 2500 q155299
## 2228 q72000
## 2486 q92899
##
## 1359 How do you feel about government-subsidized food programs (free lun
## 2142 Do you think it is okay to open old graves to get more knowledge of ancient cul
## 2419 Which type of wine would you prefer to drink outside of a me
## 2500 Would you be okay with your partner having
## 2228 Which of the following four statements best sums up how you feel about religion / faith when it
## 2486 Would you be inclined to influence a pa
## Responses
## 1359 68371
## 2142 68371
## 2419 68371
## 2500 120
## 2228 119
## 2486 101
```

```
# barplot of all questions, sorted
barplot(most_answered_q) #ugly
```



```
# create plot of comparing number of answers of most and least
barplot(answered_counts[c("q92899", "q72000", "q155299", "q34113", "q64664", "q85419")],
        main = "Frequency of Responses to \n Most and Least Answered Questions",
        xlab = "Question Numbers",
        ylab = "Answer Frequency",
        cex.names = .7,
        cex.axis = .7,
        ylim = c(0, 70000))
```



No one answered the full survey. Even when isolated to only the q columns which are the columns I understood to be questions and have options for respondents, no one completed all of those questions either. This could either be because it is such a long survey, respondents are only presented some questions, or new questions have been added and others retired thereby not exposing any one participant to truly all the questions. Because no one completed the entire survey, I attempted to answer my other questions that might lend themselves better to plots and tables. I got the most and least answered questions and made a table of the question number, the question, and amount of responses it got. I also made a bar graph of the 3 most and least answered so one could visually appreciate the difference in answered frequencies. The nature of the least answered questions surprised me because I felt there were more ‘risque’ ones in the dataset I was expecting to have been answered less.

(2.1) Your work at a social research nonprofit puts you in touch with a large, multi-year survey called the National Civic Life Survey. Each row is one respondent; columns include demographics (age, education), civic behaviors (voting frequency, volunteering hours), and attitude questions. Many questions were optional, so missing values are common. The data arrive as a Parquet file, plus a small CSV with question metadata.

Using only the tools you’ve learned so far, write out in words the steps you would take to explore this dataset

and generate a few simple plots. Do not include any R code, but you're welcome to mention R functions you might use.

First I would save both datasets into their own variable using read.csv. I would inspect both the actual data in environment to grasp the size and structure of the variables. I would also inspect the question metadata file in the environment to understand the columns of the actual data better. I would run dim, str, and names on both files to get a feel of size, structure, and naming. If any variables in the actual data set needed to be reprocessed, like changing characters to be factors I would do that. For the 4 demographic and civil behavior variables, I would get the summary to understand the range, median, and mean of the dataset. If there are any potential inconsistencies (age mean is insanely high for example) I would inspect specific rows for potential data entry errors or set thresholds for what a reasonable value could be (ex: age cannot be below 0 or above 120). I would also look into missing values of the large dataset, specifically using colSums (ex: (missing_values <- colSums(is.na(okcupid)))). I could also sort the missing values to understand which variable is missing the most entries. If I wanted to look at only completed surveys I could do complete.cases or na.omit and store these in their own variable to then analyze and plot. If I wanted to inspect any combination of answered questions (ex: only people who answered 2 wanted attitude questions and reported their age and voting frequency), I could also create these variables with subset or grep. For plotting I could make a barplot of volunteering hours to better understand the spread and frequency. I would create bins (0-2, 2-4, 4-6, 6+) and plot these on a barplot. I could also color the bins by another variable like yes/no attitude questions. I could create a scatterplot of age and voting frequency (I would need to ensure the labels on the plots explain this variable well - what does frequency mean? Is it every local election? Is it once every 4 years? What is the time this is looked at?) to understand if certain ages vote more frequently. I could create a barplot of any attitude questions with levels (ex: a question with 4 options) to understand the percentage breakdown of how everyone answered. I could also do a histogram of any attitude questions that are on a scale to show the frequency on spread of the answers.